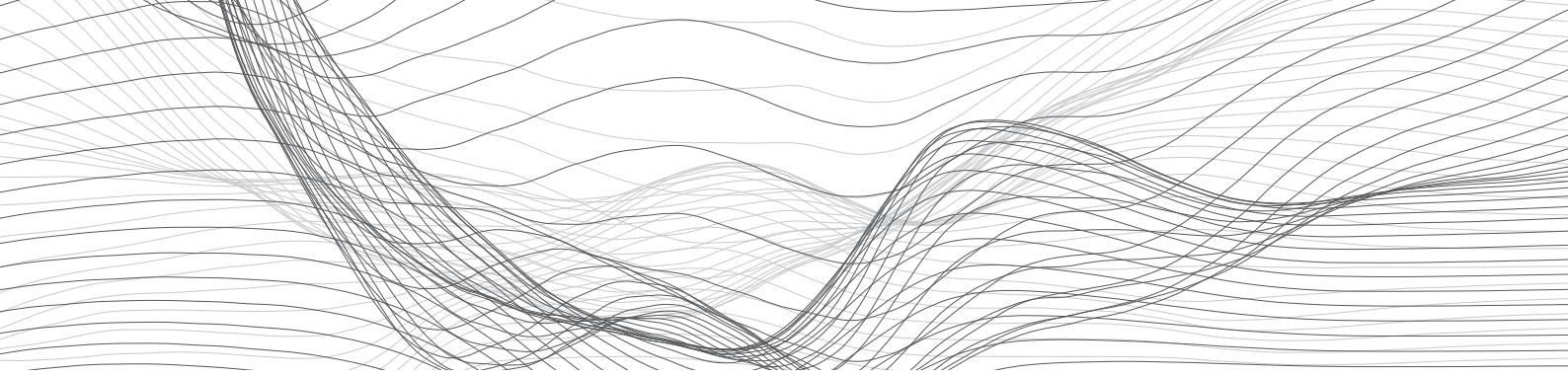


Yesterday’s controversies for tomorrow's money

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Yesterday's Controversies for Tomorrow's Money

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Abstract

While money has evolved from natural commodities to digital tokens, the debate about the ideal money has remained the same throughout the ages. Historical controversies about the origin, nature, and double-spending of money offer valuable insights for navigating the new forms of money tokens such as cryptocurrencies, central bank digital currencies and stablecoins.

JEL classification: E40, E42, E50.

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1 Introduction

Money has evolved dramatically over millennia, from commodities produced by nature to digital tokens stored in the cloud. At first glance, the novel forms of money, such as cryptocurrencies, central bank digital currencies (CBDCs) and stablecoins, appear to bear striking similarities. They all exist in the digital realm and can be transferred using a smartphone, regardless of political borders. However, apart from being digital, cryptocurrencies, CBDCs and stablecoins do not have much in common and there is an ongoing debate about which tokens are best suited to fulfilling the functions of money. There is no consensus on the best money token: while some are staunch advocates of cryptocurrencies, others argue in favour of CBDCs, and still others see the future in stablecoins. This lack of consensus should come as no surprise, as few economic phenomena are as controversial as money.

Although money has evolved alongside technology, the debate about the ideal money has a long history and the controversies have remained the same throughout the ages. When various types of money tokens are evaluated, historical controversies offer valuable insights.

This paper examines money tokens through the lens of three controversies. Section 2 considers the controversy about the origin of money. It asks whether money emerges spontaneously from human interaction and the division of labour, as the *natural law of money* maintains, or whether it is created by state decree, as the *state theory of money* contends. Section 3 discusses the controversy about the nature of money. It examines whether money is an economic good, distinct in both nature and function from credit, as the *economic good theory of money* argues, or whether money is, fundamentally, a form of credit, as maintained by the *credit theory of money*. Section 4 addresses the controversy about the double-spending of money. Since one of the defining features of cryptography is its ability to prevent the double-spending of digital tokens, this controversy remains highly relevant, particularly in the context of the modern monetary system characterised by an elastic money supply. It asks whether money deposited with a custodian should be prevented from being spent multiple times, as maintained in *classical Roman jurisprudence*, or whether deposited money may legitimately be spent multiple times, as permitted under *Anglo-Saxon common law*. Finally, Section 5 classifies the various money tokens in light of these controversies, providing a framework for understanding and navigating the diverse forms of money tokens examined

throughout this paper.

Distinction between the ontology and the condition of money

Before getting to the heart of the matter, it should be noted that the controversies are often marred by a confusion between the *ontology* and the *condition* of money. Sometimes, the discord between the protagonists suggests a divergence in their understanding of the issue at stake, with one arguing about the ontology of money and the other about its specific condition.

Controversies regarding the origin, nature or double-spending of money are not intended to describe the functioning of the current monetary system. The debate about the ontology of money is not a quest for the theory that best describes the current condition of money. Rather, it addresses the essence of money, its intrinsic meaning to human beings. The ontology of money addresses people's purpose to accept and hold money. Conversely, the condition of money describes what money is in a particular monetary system at a particular point in time. While the condition of money is determined by monetary arrangements at a specific time and location, the ontology of money is transcendent to specific monetary arrangements.

2 Controversy about the origin of money

> *Money is the result of human action but not the result of human design.*
>
> Friedrich von Hayek

> *Money is peculiarly a creation of the State.*
>
> John Maynard Keynes

Why is money valuable and how does it come into existence? The debate about the origin of money addresses the problem of circularity between the value and acceptance of money: people accept money because it is valuable,

and money is valuable because people accept it. Explaining the value of money by referencing its acceptance and the acceptance of money by referencing its value is a circular argument. Saying that an economic agent accepts something as a medium of exchange because other agents also accept it creates a recursive or coordination problem. How would agents coordinate their choice of what would circulate as a common medium of exchange?

The origin of money is a bone of contention among economists, splitting them into two camps as to the role of the state in the existence of money. The *natural law of money* maintains that money emerged spontaneously and progressively over the ages from the market exchange process. In contrast, the *state theory of money* maintains that money is a creation of the state and derives its value from being issued by the state. Both theories claim to have empirical evidence in their favour.

2.1 Natural law of money

The natural law of money, notably promulgated by Menger (1871) and Brough (1896), asserts that money arose spontaneously through market exchanges, thereby solving the problem of the double coincidence of wants that characterises barter. As humanity has evolved from a barter economy to a monetary economy, various commodities were selected to fulfil the role of money as a generally accepted medium of exchange. Primitive forms of money, such as salt and spices, were gradually replaced by metallic money. Cruder forms of metallic money, such as copper and bronze, were also gradually replaced by silver and gold. As money evolved to meet the needs of trade, silver and gold became the dominant monetary metals of the most advanced nations.

Regression theorem by Mises

What is the solution proposed by the natural law of money to the alleged problem of circularity between the value and acceptance of money?

The regression theorem formulated by Mises (1912) states that people began to use and accept commodities as media of exchange for their nonmonetary use value. This theorem circumvents the problem of circularity by tracing the value and acceptance of money back to the nonmonetary value of the commodities before they began to be used as money. As Mises (1949, p. 406) stated:

> The regression theorem aims at interpreting the first emergence of a monetary demand for a good which previously had been demanded exclusively for industrial purposes as influenced by the exchange value that was ascribed to it at this moment on account for its nonmonetary services only.

The nonmonetary value of the commodities used as money throughout history solves the problem of coordination. If a commodity has a nonmonetary use, an economic agent will accept that commodity as a medium of exchange not because all other agents accept it as a medium of exchange, but because it has a nonmonetary value. The transition from a barter economy to a monetary economy was gradual, as the commodities originally used as money also had nonmonetary uses. It is impossible to pinpoint exactly when a barter economy became a monetary economy, since the goods used as money were also used for consumption or production. If a weaver exchanges a blanket for salt, she does not need to know in advance whether she will consume the salt herself or exchange it for wool from a shepherd. The good used as money is in demand for both its nonmonetary and monetary uses. Of course, if people increase their monetary demand for the good used as money, its price may rise above the level that would be reached if its value were determined solely by its nonmonetary demand. Mises (1949, p. 407) clarified that "[the regression theorem] certainly does not involve explaining the specific monetary exchange value of a medium of exchange on the ground of its industrial exchange value." In other words, although the regression theorem states that people began to use a good as a medium of exchange because it had a nonmonetary value, this does not mean that the value of the monetary good is independent of its monetary demand. The value of the good used as money increases with its monetary demand. Nevertheless, it is the nonmonetary value of the good that initially prompted its use as a medium of exchange.

The regression theorem does not enter into the circular argument because the reason why people began to accept money can be traced back to the nonmonetary value of the goods used as money. It is true that people accept money because of its expected purchasing power and that its purchasing power is determined by people's acceptance of money. However, the expected purchasing power of money is determined by its recent purchasing power. People evaluate the future purchasing power of money on the basis of their past experience. They accept money today because they know that it had value yesterday. And

people accepted money yesterday because they knew that it had value the day before, and so forth. Mises (1949, p. 406) stated that the reasoning can be regressed back to the beginning of money as follows:

> If we trace the purchasing power of money back step by step, we finally arrive at the point at which the service of the good concerned as a medium of exchange begins. At this point yesterday's exchange value is exclusively determined by the nonmonetary – industrial – demand which is displayed only by those who want to use this good for other employments than that of a medium of exchange.

The regression theorem also explains how the substrate of money can evolve over time, as highlighted by Simmel (1907). For instance, the theorem rationalises why a banknote that can be converted into precious metals retains its immediate exchange value when the obligation to redeem it is removed and the banknote becomes fiat currency. The removal of the requirement to redeem a banknote or a bank deposit in precious metals represents a fundamental change in the nature of the banknote or deposit. According to the *rational expectation hypothesis*, the value of money would decrease significantly and immediately following the decision to remove the convertibility of banknotes into precious metals. However, people typically continue to assess its purchasing power on the basis of their recent experience. For some time, they continue to accept the unredeemable notes under the same conditions as the redeemable notes. Once the banknotes have been accepted as a common medium of exchange, people continue to use them for convenience, despite the abolition of the obligation to redeem them. The purchasing power of the redeemable banknotes is transferred to the – physically identical – unredeemable banknotes. Of course, the abolition of the redemption obligation is generally accompanied by a gradual increase in the volume of banknotes or deposits issued. This will result in a gradual erosion of their purchasing power over time, as people adjust their assessment of the exchange value of money on the market day after day.

Role of the state in the natural law of money

While the natural law of money holds that money arises independently of the state, it does not ignore that the state has increased its influence on monetary matters over the ages. Brough (1898) asserted that "money was circulating everywhere in the world before men even thought of making laws for its reg-

ulation." Menger (1871) recognised that the state has taken on two roles in monetary matters: defining the standard of value and manipulating the standard of value. He emphasised the significant difference between these two roles, offering contrasting assessments of them.

Defining the standard of value

The first role of the state was to define the standard of value in terms of the weight and fineness of a specific precious metal. This role does not contradict the natural law of money. Menger (1871, p. 261-262) summarised the role of the state in the emergence of money as follows:

> The origin of money (as distinct from coin, which is only one variety of money) is, as we have seen, entirely natural and thus displays legislative influence only in the rarest instances. Money is not an invention of the state. It is not the product of a legislative act. Even the sanction of political authority is not necessary for its existence. Certain commodities came to be money quite naturally, as the result of economic relationships that were independent of the power of the state.
>
> But, if in response to the needs of trade, a good receives the sanction of the state as money, the result will be that not only every payment to the state itself but all other payments not explicitly contracted for in other goods can be required or offered, with legally binding effect, only in units of that good. ... Thus, the sanction of the state gives a particular good the attribute of being a universal substitute in exchange, and although the state is not responsible for the existence of the money-character of the good, it is responsible for a significant improvement of its money-character.

The state did not arbitrarily choose to mint coins of silver or gold. These metals were chosen because of their high exchange value. This indicates that precious metals were in high demand for monetary purposes prior to the state beginning to mint coins. At some point, the state began minting precious metal coins that are easy to handle. The state's stamp certifies that the coin contains the required weight and fineness of the precious metal. Given the difficult task of weighing and testing metal, the establishment of a standard of value by the state was a valuable service for citizens. This relieves the public's need to

check the weight and fineness of precious metals at each transaction, thereby increasing their acceptance. Menger (1871, p. 282) reported the following:

> The best guarantee of the full weight and assured fineness of coins can, in the nature of the case, be given by the government itself, since it is known to and recognised by everyone and has the power to prevent and punish crimes against the coinage. Governments have therefore usually accepted the obligation of stamping the coins necessary for trade.

Manipulating the standard of value

The second role taken on by the state has been to manipulate the standard of value. Having acknowledged the benevolent role of defining the standard of value, Menger (1871, p. 282-283) deplored the manipulating behaviour of the states, who

> ... have so often and so greatly misused their power that economising individuals eventually almost forgot the fact that a coin is nothing but a piece of precious metal of fixed fineness and weight, for which fineness and full weight the honesty and rectitude of the mint constitute a guarantee. Doubts even arose as to whether money is a commodity at all. Indeed, it was finally declared to be something entirely imaginary resting solely on human convenience. The fact that governments treated money as if it actually had been merely the product of the convenience of men in general and of their legislative whims in particular contributed therefore in no small degree to furthering errors about the nature of money.

Manipulating the standard of value conflicts with the natural law of money. Such manipulations have taken the form of debasement or the enforcement of fiat money through legal tender laws and have given rise to monetary phenomena associated with Gresham's law.

Debasement The earliest forms of monetary manipulation involved debasing coins by reducing their precious metal content. The public might accept these debased coins at the same value as the originals – either because the reduction in precious metal went unnoticed or because legal tender laws required

that debased coins be accepted on the same terms as the original coins. Debasement benefits the state at the expense of the people, with the gain equal to the increase in the quantity of coins in circulation. For example, the state might withdraw 100 coins, each containing 10 grams of gold, from circulation. By melting these down, it can issue 200 new coins, each with the same face value as the originals but containing only 5 grams of gold and 5 grams of a cheaper metal, such as copper. The state effectively generates revenue from this process equal to the face value of the additional coins created minus any minting costs.

Fiat money One extreme form of debasement is the introduction of fiat money, which has no intrinsic value[1] and can therefore be issued without economic constraints. Fiat money has been issued through legal tender laws, enabling states to enforce the acceptance of unredeemable banknotes in payment. In the words of Sennholz (1985), the law of legal tender gives rise to "compulsory money." The introduction of fiat money prompts the question of whether contracts concluded beforehand should be honoured in precious metals or fiat money. Do the dollars mentioned in the contracts refer to gold dollars or paper dollars? Legal tender laws stipulate that contracts must be interpreted as referring to paper dollars, thereby mandating the acceptance of fiat money over metallic money (Goldberg, 2009). The shift from metallic money to fiat money has always been decreed by the state, suspending the obligation to redeem banknotes and deposits in metallic money.

Such a step has typically been taken under the pressure of crises or wars. In the United States, for example, legal tender was imposed by law in times of national emergencies to finance the revolution between 1775 and 1779. By 1780, the people refused to accept the Continental dollar, as its value had fallen to three cents, eventually forcing the state to repeal its legal tender laws. Later, in 1862, President Lincoln signed the Legal Tender Act to help finance the civil war. The Act authorised the issuance of United States notes, declared to be "lawful money and legal tender in payment of all debts, public and private, within the United States ..."

Gresham's law Manipulation of the standard of value has also given rise to the peculiar monetary phenomenon known as Gresham's law, which states

[1] A money has an intrinsic value if its substrate has a value beyond its monetary use.

that bad money drives good money out of circulation. As Brough (1898, p. 36) observed, the legal tender reverses the natural law of money:

> The more efficient money will always drive from the circulation the less efficient if the individuals who handle money are left free to act in their own interest. It is only when bad money is endowed by the state with the property of legal tender that it can drive good money from the circulation.

If left free, the exchange value of different coins circulating simultaneously in the economy reflects the exchange value of the different metals in the bullion market and their relative minting costs. For example, if the price of silver increases relative to the price of gold in the bullion market, the exchange value of silver coins will increase in relation to that of gold coins. Alternatively, given that the cost of minting a coin is not proportional to its weight, smaller coins will have a higher exchange value relative to their weight than larger coins.

Whenever the state sets an official fixed exchange ratio between coins of different metals, this ratio will soon or later deviate from the exchange ratio that would have formed freely, creating arbitrage opportunities. For instance, if the state sets an official exchange ratio between gold and silver coins of 15 to 1 and the international bullion market quotes the metals in a ratio of 16 to 1, arbitrageurs will buy gold for silver at the official ratio in the domestic market and sell it at the free international market ratio. As a result, the undervalued (good) money will leave the domestic market for the international market, while the overvalued (bad) money will flow into the domestic market. This phenomenon was observed by Sir Thomas Gresham (1519-1579), after whom the law was named. Rothbard (1962, p. 1080) summarised the law as follows: "Actually, Gresham's Law should read: Money overvalued by the state will drive money undervalued by the state out of circulation."

The same mechanism occurs when the state imposes an exchange ratio between large and small coins on the basis of their weight of precious metal, without accounting for minting costs. This in turn discourages mints from minting small coins and explains the intermittent shortages of small denomination coins observed throughout history and documented by Sargent and Velde (2002).

In summary, while the natural law of money acknowledges the state's growing involvement in monetary matters, it maintains that state intervention is not essential for money to come into existence. It offers a critical perspective

on this development, raising concerns about the possible misuse of coercive power against the people. Against this backdrop, the significance of the central bank's independence from political influence becomes apparent. If governments did not have a long tradition of taking advantage of their power in monetary matters, the independence of central banks would be superfluous.

2.2 State theory of money

The competing theory to the natural law of money is the state theory of money, according to which "money is peculiarly a creation of the State", as noted by Keynes (1930, p. 4). The state theory of money, also called the chartalist theory of money, can be traced back to Plato[2] and was developed by Knapp (1905) and endorsed by Michell-Innes (1914), Keynes (1930), and Goodhart (1998).

The state theory of money asserts that the coordinating role of the state was instrumental in the emergence of money. The problem of circularity between value and acceptance of money is solved by the state declaring what is to be accepted as payment for obligations, particularly taxes. As Peacock (2013, p. 45) claimed:

> The state theory circumvents the circularity of positing that people accept money for the reason that money is widely accepted because the theory specifies a first mover, viz. the state, that guarantees that it will accept the money it issues as the means with which taxes are to be paid.

Without state intervention, money would not exist. Money began to derive its value from being accepted as a payment for taxes, not from its intrinsic value. Writing during the gold standard era, Michell-Innes (1914, p. 152) asserted that:

> It is the tax which imparts to the obligation its 'value'. A dollar of money is a dollar, not because of the material of which it is made, but because of the dollar of tax which is imposed to redeem it.

People demand and accept money in market exchanges because they know that the state will accept it as a payment for taxes. The demand for money

[2]In *The Republic*, Plato denounced the use of precious metal as money because it hinders economic regulation of the *polis* by the government. He thus called for a government fiat currency.

stems from tax obligations, which require people to acquire money through market exchanges to pay their taxes. Peacock (2013, p. 39) maintained that:

> The power of the State to impose a tax or similar liability payable in its own currency is sufficient to create a demand for that currency and give it value.

While the state need not be the sole issuer of money, the state's acceptance of something as a means of payment is essential to the emergence of money. It is the state's acceptance of something as money that determines what money is. Knapp (1905, p. 95) explained that "all means by which a payment can be made to the State form part of the monetary system."

The state theory of money turns the relationship between money and the market on its head. Recall that the natural law of money posits that money emerged spontaneously from the process of market exchange to solve the problem of the double coincidence of wants. The emergence of money, therefore, required the preexistence of economic exchanges, and money is perceived as enhancing exchanges and the division of labour. The state theory of money reverses this relationship. It posits that human beings began to engage in economic exchanges not for their own interest, but rather to acquire the money necessary to settle their tax obligations, as exposed by Hoppe and Langton (1994). Peacock (2013, p. 37, 41) claimed that:

> ... in the absence of taxes to be paid in coin, peasants or other social groups might lack any incentive to turn to the market. ... The peasantry engaged in commercial activity only insofar as it was constrained to do so by the fiscal pressures of the state.

In other words, the state theory posits that human beings did not spontaneously engage in economic exchange to enhance their welfare through the diversification of consumption and the division of labour but were forced to exchange their production with each other to acquire money to pay taxes. While the natural law views money as a means of facilitating economic exchange, the state theory views economic exchange as a means of circulating money and paying taxes. While the natural law views taxes as a hindrance to economic exchange, the state theory views taxes as the driver of economic exchange. Economic exchange is not seen as voluntary and mutually beneficial for humanity but as a duty imposed by the state. Disagreeing with this view, Mises (1912, p. 511) summarised that

> [the etatist school] undertook the attempt to represent all the phenomena of social life merely as emanation of the exercise of power by princes and others in authority. It is only a logical extension of its doctrine to endeavour eventually to represent money also as being created merely by force.

Despite the criticism it has attracted, Knapp's state theory of money achieved remarkable influence. This was due not only to Knapp's talent for conceptualisation, exposition, and persuasion, but also to the favourable intellectual and political context in which it emerged. As Schumpeter (1954, p. 1091) observed, "many people and especially politicians at that time welcomed a theory that seemed to offer a basis for the growing popularity of state-managed money ..."

2.3 Empirical evidence of the origin of money

The natural law and the state theory of money have opposing views on the origin of the value of money. Interestingly, both the advocates of the natural law and the advocates of the state theory claim that their doctrine is supported by empirical evidence. We discuss this evidence in more detail below.

Historical evidence for the natural law of money

The empirical support for the natural law of money is twofold. First, it is based on the historical observation that money has emerged and circulated independently of state jurisdiction. The natural law of money explains the spontaneous emergence of cocoa beans as money in the Aztec economy or of mollusc shells in the American Great Plains (Einzig, 1966). The archaeological studies referenced in Newman (2025) and Thompson (2003) provide substantial evidence supporting the hypothesis that precious metals functioned as a medium of exchange long before states began minting coins. Silver was used as money in ancient Mesopotamia in the form of unminted ingots or jewellery for thousands of years prior to the introduction of coins. While it may be true that coins have always been minted by the state or the ruler, precious metals did not start circulating as money in the form of coins. Moreover, several national monies were widely circulated outside their respective state jurisdictions. The Florentine guilder and the Venetian ducat, for instance, were widely accepted

as a medium of exchange throughout Europe during the Renaissance. Similarly, the British pound sterling and the US dollar have been accepted as money in many foreign countries without the intervention of the local authorities, and often even against their will. The recent emergence of cryptocurrencies such as Bitcoin is another case in point. These examples demonstrate that private agents can spontaneously choose a medium of exchange independent of the state.

The second piece of empirical support for the natural law of money comes from the demonetisation of gold. The natural law and the state theory make opposite predictions about the effect of demonetising gold on its price. According to the natural law, the demonetisation of gold would not result in a reduction in its value; in contrast, it would lead to an increase in its value if demonetising gold paved the way for more fiat money to be issued. Since people had chosen gold as money spontaneously, they would continue to demand it for its monetary function, even after the state had demonetised it. In contrast, proponents of the state theory claimed that demonetising gold would reduce its price because it derives its value from being accepted as money by the state. As asserted by Michell-Innes (1914):

> ... if gold were left to be governed by the ordinary laws of commerce [i.e., if gold ceases to be used as money], there can be no question but that the price would fall, to the great loss of shareholders in gold mines and the great benefit of the rest of humanity.

The demonetisation of gold in the second half of the twentieth century provides a unique opportunity to test these opposing predictions. Rothbard (1962, p. 10) summarised the outcome of the experiment as follows:

> There can be no genuine laboratory experiments in human affairs, but we came as close as we ever will in 1968, and still more definitively in 1971. Here were two firm and opposing sets of predictions: the [natural law], which stated that if the dollar and gold were cut loose, the price of gold in ever-more inflated dollars would zoom upwards; and the [state theory] ... maintaining that the price of gold would, if cut free, plummet from $35 to $6 an ounce. ... What then of the laboratory experiment? Flouting all the predictions of the [state theory], there was no contest as between it and the [natural law]: not once did the price of gold

> on the free market fall below $35. Indeed it kept rising steadily, and after 1971, it vaulted upwards, far beyond the once seemingly absurdly high price of $70 an ounce. Here was a clear-cut case where the [natural law] forecasts were proven gloriously and spectacularly correct, while the [state theory] predictions proved to be spectacularly wrong.

Historical evidence for the state theory of money

Proponents of the state theory of money claim that empirical facts support their view that money is a creation of the state. They observe that money is primarily associated with the state. Currently, there is almost a one-to-one relationship between currencies and states. As Goodhart (1998) claimed, the state theory of money seems to fit the current situation of state-issued currencies better than the natural law of money does. Goodhart (1998, p. 417) argued the following:

> Even if one should accept the [natural law] theory of the evolution of metallic coins as money, it is problematic to use that same theory in its pure form to explain why agents should suddenly *all* be willing to jump from using paper notes which were ultimately claims on precious metals (i.e., private or public sector bank notes convertible into such precious metals) to paper notes which were backed by no specific assets. Instead those notes were, and are, backed by the power of government (e.g., legal tender laws) and its ability to impose taxes payable (and often only payable) in that fiat currency (as well as legal tender for the discharge of all other payments within the country). Thus, the [natural law] theory has difficulties with explaining the introduction and use of fiat currency. The [state] theory has no such difficulties. The transition was entirely natural.

Behind their apparent disagreement, the two theories agree on one point: the central role of the state in the emergence of fiat money. As we saw above, the natural law acknowledges that legal tender laws are necessary for the transition from banknotes convertible into precious metals to nonconvertible banknotes. The natural law does not claim that this transition was the spontaneous result of market forces. It acknowledges that fiat money is a creature

of the state. Here, we encounter a striking example of Goodhart's conflation of the ontology of money with its contingent condition. While natural law acknowledges that money's present condition is the product of state intervention, it does not take the existing monetary system as its object of explanation. Its concern is instead with identifying the essence of money.

However, the two theories disagree on what gives fiat money its value. According to the natural law and the regression theorem, the value of fiat money derives from experience gained in past transactions, with current market conditions progressively adjusting that value. The transition from convertible to nonconvertible banknotes is therefore considered a seamless process that has had little impact on the way economic agents form expectations about the value of money on the basis of their recent experience. The value of money realised in yesterday's transactions forms the prior beliefs with which agents enter today's market. The value of money does not result from the possibility of using it to pay taxes. However, the fiscal stance can influence the state's tendency to finance its budget through money creation, which in turn affects the value of the money. If there is a trade-off between tax collection and money creation to finance the state, an increase in tax collection combined with a decrease in money creation strengthens the value of money, whereas the reverse would weaken it. However, it is not the acceptance of fiat money for paying taxes that gives money its value. Money would not lose its value if the state were to stop collecting taxes and instead finance its budget with the proceeds from natural or mining resources. In contrast, the state theory asserts that the value of fiat money derives from its ability to pay taxes.

Do cryptocurrencies invalidate the regression theorem?

The natural law and the state theory of money offer contrasting perspectives on the emergence of cryptocurrencies. While it is clear that cryptocurrencies are not created by the state, the two theories differ in their explanations of the source of the value of these currencies.

From the perspective of state theory, the lack of fiscal backing is seen as a weakness of cryptocurrencies and leads to them being regarded as nothing more than a speculative bubble. For example, Buiter (2014) and Shiller (2019) argued that the valuation of Bitcoin is sustained largely by a contagious social narrative that fuels speculative demand rather than by underlying economic fundamentals. From this perspective, cryptocurrencies – and, in Buiter's anal-

ysis, gold as well – exhibit the defining characteristics of a speculative bubble. However, this interpretation presupposes that economic fundamentals determine the objective value of a good, thereby overlooking the inherently subjective nature of value. Nevertheless, value is not an objective property of goods. Rather, it emerges from individuals' subjective preferences, implying that no good has an objective or intrinsic value determined by economic fundamentals and independent of human valuation. Compared with any other economic goods, gold and cryptocurrencies are no less objectively valuable.

The emergence of cryptocurrencies also raises the question of whether the regression theorem can adequately account for their origin and whether cryptocurrencies pose a challenge to the theorem itself. The theorem holds that the value of money can be traced backwards through a regression process to the nonmonetary value of the first commodities adopted as a medium of exchange. Since cryptocurrencies lack apparent nonmonetary use value, one might argue that they invalidate the theorem by demonstrating that money can emerge without being initially grounded in a good with nonmonetary value. For example, Suede (2011) argued that an object does not necessarily need to be valued in direct exchanges to become accepted in indirect exchanges. Rather, the perceived benefits of using the object as a medium of exchange may be sufficient to lead individuals to adopt it for this purpose. He therefore concluded that the regression theorem provides an incorrect account of the origin of money.

There are, however, two lines of argument that support the compatibility of the theorem with cryptocurrencies. The first argues that Bitcoin initially had a value – albeit a very low one – but a value that was nonmonetary in nature. According to this view, the first individuals who spent a few cents to acquire Bitcoins did so because they valued Bitcoin itself, not because they anticipated the future monetary demand for it. It is important to remember that the value of Bitcoin remained close to zero for several years after its launch, before gradually increasing as demand grew. Those who first acquired Bitcoins for a small fraction of a dollar were not primarily motivated by the expectation of future profits; rather, they acquired them because they appreciated Bitcoin in itself.[3] The reasons for acquiring Bitcoin were likely as diverse as they were subjective. Just as the lustre and beauty of gold have long captivated humankind,

[3]To illustrate, consider the first documented purchase of real-world goods using Bitcoin. On May 22, 2010, Laszlo Hanyecz purchased two pizzas for 10,000 Bitcoins, which were worth approximately $40 at the time.

the elegance of Bitcoin's underlying algorithm has fascinated cryptography enthusiasts. While some people spend money on virtual games or on items such as Pokémon cards, these early adopters were willing to incur small costs – such as a few dollars' worth of electricity – to mine Bitcoins and participate in the experiment. Gertchev (2023) explained the following:

> It could then be safely assumed that Bitcoins served the purpose of conveying a specific antiestablishment worldview. The first demand factor ... was rooted in their capacity to project a certain point of view. In a sense, Bitcoins were comparable to an artistic medium of expression, such as music, literature, and painting. Thanks to that initial source of value, Bitcoins had a reference point that positioned them relative to other goods and services. From there onwards, the technological feature that characterize them led to an expansion of their demand.

The second line of argument defending the compatibility of cryptocurrencies with the regression theorem holds that the theorem is intended to explain the initial emergence of a medium of exchange from a barter economy rather than the emergence of new media of exchange within an already existing monetary system. Davidson and Block (2015) claimed that the theorem does not apply to Bitcoin because Bitcoin did not emerge in a barter economy. Instead, it appeared within a developed monetary system in which individuals already used established currencies, such as the dollar, to express prices and evaluate goods. Consequently, Bitcoin did not need to transition from a nonmonetary commodity into money through a process of indirect exchange. Its initial exchange value could be determined through direct comparisons with existing money prices. This argument is consistent with the observation by Simmel (1907) that money develops within an already existing network of exchange relations and prices, allowing new monetary forms to derive their value from the established monetary order rather than from an independent nonmonetary use.

One may venture a speculative hypothesis regarding the indirect role of the state in the emergence of cryptocurrencies. It is worth asking whether a significant monetary demand for cryptocurrencies would have arisen in the absence of state-issued fiat money. Faced with a choice between fiat money, whose supply is ultimately subject to political decisions, and a digital currency governed by a predetermined cryptographic protocol (Selgin, 2015), some individuals

may reasonably decide to exchange a portion of their holdings of the former for the latter. Whether the same demand would have emerged in a monetary system based on precious metals is much less clear. If individuals had already enjoyed a relatively stable, market-based monetary system, the incentives to adopt cryptocurrencies as an alternative store of value or medium of exchange might have been considerably weaker. In this sense, one may speculate that the rise of cryptocurrencies was, paradoxically, an unintended creation of the state. Their emergence as a monetary phenomenon may therefore have been conditional on the existence of state-issued fiat money.

3 Controversy about the nature of money

> *Gold is money. Everything else is credit.*
>
> John Pierpont Morgan

> *Credit and credit alone is money.*
>
> Alfred Mitchell-Innes

Having discussed the controversies surrounding the origin of money, let us now turn our attention to the question of its nature. Two competing conceptions of money can be distinguished. The first considers money as an economic good, the second as a credit. The *economic good theory* posits that money and credit have different purposes and natures. Conversely, the *credit theory* posits that money is, in essence, a form of credit.[4]

Although most proponents of the credit theory also subscribe to the state theory of money discussed in the previous section, the question of the nature of money is logically distinct from the question of its origin.

[4]Schumpeter (1954, part III, chap. 7.4) offered a historical reconstruction of the emergence of these theories alongside the development of banking. His account, however, was genealogical rather than ontological. It traced how conceptions of money evolved with changing banking practices, but it did not investigate the nature of money itself. Our concern in what follows is therefore not with the historical evolution of these theories but with the ontological question of what money is.

3.1 Economic good theory of money

The economic good theory of money is based on the Aristotelian premise that money is a medium of exchange for acquiring goods and services.[5] This view was taken up by Jean-Baptiste Say and the British Currency School, and was established later by Menger (1871), Jevons (1875) and Mises (1912).

According to this theory, money is something commonly accepted as a medium of exchange. The primary function of money is to serve as a common medium of exchange, and the other functions – as a store of value and a unit of account – are derived from this. As money can be exchanged immediately for other goods, it also serves as a store of value. Since all goods are exchanged for money, money is used as a unit of account to measure their value.

Economic goods are divided into three categories: consumption goods, production (capital) goods, and money. Money can be spent to acquire consumption goods or production goods, but money is neither a consumption nor a production good. Money is a category of its own. The division of economic goods into consumption goods, production goods, and money is based on the use and subjective valuation of goods made by human beings. Although a good's physical characteristics significantly influence its utilisation, it is ultimately the purpose followed by human beings that determines its utilisation. For instance, gold can be used as a consumption good in gold-leaf mosaics, as a capital good in the form of gold-plated conductors for electronics, or as a medium of exchange in the form of coins. Tobacco is a consumption good for smokers and a production good for beekeepers and can also be a medium of exchange for prisoners. The subjective purpose of human beings, rather than the objective characteristics of goods, is to divide economic goods into the consumption–capital–money categories.

The economic good theory of money has historically been equated to the commodity or metallist theory of money. This can be traced back to the pivotal role played by commodities such as salt or spices, as well as precious metals, throughout history. Nevertheless, the understanding of money as an economic good is not limited to its physical properties. The form or substrate of money can evolve from concrete to more abstract, as described by Simmel (1907) .

[5] In the *Nicomachean Ethics* (book 5, chapter 5), Aristotle stated: "And for the future exchange – that if we do not need a thing now we shall have it if ever we do need it – money is as it were our surety; for it must be possible for us to get what we want by bringing the money. Now the same thing happens to money itself as to goods – it is not always worth the same; yet it tends to be steadier."

Intangible things with no intrinsic value used as a medium of exchange, such as fiat money or cryptocurrencies, also comply with the economic good theory of money.

Chinese wall between money and credit

The economic good theory of money holds that money differs, by nature, from credit. Whereas money is a good commonly accepted as a medium of exchange, credit refers to an *intertemporal* transaction in which goods are exchanged at different points in time. Credit transactions do not require the existence of money and predate the advent of money in barter economies. Indeed, exchanges over time can occur in a barter economy, where goods are delivered and received at different points in time. A barter economy can be understood as a system of reciprocal credit relationships between individuals rather than as a marketplace characterised by the simultaneous exchange of goods. For example, the service of ploughing with oxen in spring could be exchanged for wheat flour at harvest time. In a monetary economy, a credit transaction occurs when a payment is made in money subsequent to the delivery of goods or when money is lent and borrowed.

Money and credit serve different purposes for human beings because life on earth is characterised by scarcity, the passage of time, and uncertainty. The risky nature of life means that money and credit belong to different economic categories. Howden (2023, p. 164) observed that "uncertainty is the root of all money" and that "the only way to hedge against [fundamental or Knightian uncertainty] is to hold an asset that pays out (can be exchanged) at any time and at a predefined value. Money is the unique financial asset that enables such an exchange." The purpose of human economic activity is to alleviate scarcity and reduce uncertainty. Building reserves, protecting against adversity and taking risks are all economic categories essential to human life, and controlling goods plays a pivotal role in the economic process. In this context, the holding of money and the granting of credit fall into two distinct categories of human purpose. The purpose of holding money is to control present goods and to reduce uncertainty because holding money embodies freedom of action. In contrast, the purpose of granting credit is to transfer control over present goods to the borrower. The process of granting credit has the effect of reducing the lender's freedom of action and increasing the level of uncertainty. Holding money and granting credit have opposite purposes.

When we consider money as an economic good, it means that money does not represent a claim on goods against anyone. As a medium of exchange, money can be exchanged on the market for other goods. However, its purchasing power does not derive from any claim on goods that money would embody. This is distinct from the valuation of a credit paper. For example, a bond that pays a specific sum of money at a future date derives its value today from the claim to future money that it embodies. However, since money does not embody a claim on anything, its value does not derive from a promise to be fulfilled by someone. As Howden (2023, p. 166) emphasised, "[money] is not just one extreme of a liquidity or value spectrum, but actually defines the spectrums." Its purchasing power is derived from the relative valuation – or relative supply and demand – of money and other goods in everyday market transactions. Each transaction contributes to the process of determining the value of money in relation to other economic goods.

3.2 Credit theory of money

The credit theory of money rejects the distinction between money and credit, and states that money is a credit. This theory was proposed by Macleod (1889), Solvay (1900), Michell-Innes (1914), Keynes (1930), and more recently by Ingham (2004), Martin (2013), and Peacock (2013).

Money is seen as an abstract measure of claims on economic goods rather than an economic good that is exchanged to settle transactions. Macleod (1889) was the first economist to formulate a theory in which money was equated with credit. While addressing the distinction between money and credit, he stated that (p. 82):

> Money and Credit are essentially of the same nature: Money being only the highest and most general form of Credit. They are each a Right or Title to demand something to be paid or done by some one else.

Influenced by him, Michell-Innes (1914) promoted the credit theory of money in the 20th century, a theory he summarised as follows:

> There is no such thing as a medium of exchange. A sale and purchase is the exchange of a commodity for a credit. Credit and credit alone is money. The monetary unit is an abstract standard for the measurement of credit and debt.

Mitchell-Innes continued as follows: "The eye has never seen, nor the hand touched a dollar," just as the eye has never seen a metre and the hand has never touched a gramme. Although we can observe distances of one metre and handle objects weighing one gramme, the metre and the gramme themselves are abstract units of measurement rather than objects. By the same reasoning, a monetary unit such as the dollar is an abstract unit of account that measures credit rather than an identifiable and tradable economic good. From this perspective, the primary function of money is to serve as an abstract unit of account. Indeed, Ingham (2004) argued that money's role as a unit of account predates – and does not depend upon – the existence of a medium of exchange.

The credit theory considers money a claim on goods that producers acquire by selling their production. As Michell-Innes (1913) noted:

> By buying we become debtors and by selling we become creditors, and being all both buyers and sellers we are all debtors and creditors.

When producers sell goods in exchange for money, they acquire credit claims that record their contribution to total production and give them the right to consume the same amount. The accumulation of money resulting from the sale increases the credit of sellers and their right to dispose of goods within the economy. Conversely, the decrease in money held by buyers following the purchase means that their credit has been reduced.

Clearly, the credit theory of money has a different interpretation of the meaning of a credit transaction than that of the economic goods theory does. In the economic good theory of money, the dividing line between money and credit is determined by the issue of *control* over present goods. While the holder of money retains control over present goods, the lender cedes control to the borrower. For instance, a sale that is paid for with money is not considered a credit sale, since the seller takes control of the sale proceeds immediately upon receiving the money. However, when the seller sells on credit, he enters into a credit transaction because he relinquishes immediate control over present goods in exchange for a claim on future money, i.e., future goods.

In contrast, the credit theory of money focuses on the *spending* on present consumption or production goods. Since holding money corresponds to a lack of spending on consumption or production goods, i.e., it is an act of saving, it falls, for the credit theory, into the same category of human purposes as granting credit. Holders of money are said to grant credit because, while

they hold their money, they postpone spending on consumption or production goods, just like bondholders do. Money is thus seen as a credit that can be redeemed on demand for consumption or production goods, which differs from long-term credit only in terms of its maturity, not its nature. Forms of credit vary in their degree of *moneyness*, which captures their degree of acceptance for transaction settlement.

3.3 Assessing the economic good and the credit theories of money

Does the economic good theory or the credit theory of money capture the correct nature of money? Remember that the question is not which theory best describes the current monetary system. Obviously, the credit theory of money provides a more accurate description of the current condition of money under fractional-reserve banking. The question is rather which theory better captures the purpose of money for humans. Do people pursue the same purpose by granting credit as they do by holding money? Do money and credit belong to the same economic category in the human mind?

The intellectual exercise of equating money with credit in the credit theory can be understood by the fact that holding money and granting credit both reflect an act of saving. This point is well taken. Nevertheless, since life on earth is characterised by scarcity, uncertainty and the passage of time, it may be misleading to confuse holding money – an action that reduces risk and increases flexibility – with granting credit – an action that increases risk and reduces flexibility. *Controlling* present goods with money is not the same as *ceding control* of present goods to borrowers through credit. The credit theory is a bit like confusing salt and sugar because they look so similar. Yes, both salt and sugar are white condiments used in cooking, but they serve different purposes, and failing to recognise this can have unfortunate consequences for the meal served.

These competing theories provide nuanced accounts of the credit-based origin of money in primitive societies, the rate-of-return dominance puzzle of money, and the challenge of establishing monetary uniformity.

Debate about the credit-based origin of money in ancient Near Eastern empires

The economic good theory and the credit theory of money lead to different interpretations of how primitive and ancient societies should be classified. It is widely accepted that economic and social relations predate the emergence of a medium of exchange. Even in primitive tribal societies, some degree of division of labour existed, giving rise to patterns of production, exchange, and mutual obligation. Later, the ancient empires of Mesopotamia and Egypt developed administrative institutions that collected taxes and rents from peasants in the form of grain, textiles, and labour services. These resources were then redistributed to administrators, soldiers, and other workers. Managing this system required an accounting framework to record the resources received and disbursed, typically expressed in units such as barley rations or days of labour. These basic historical facts are largely uncontested. What remains contested is their interpretation and, in particular, what they imply about the origin and nature of money.

According to the economic good theory of money, keeping accounts of barley rations does not turn a barter economy into a monetary economy. When the authorities record that a farmer has delivered the required quantity of barley as payment of taxes, the accounting entry merely documents the transfer of goods and does not itself constitute a monetary transaction. Similarly, debt and credit relationships that arise between individuals in a barter economy are not regarded as monetary phenomena because they do not involve the use of a medium of exchange. This, however, does not preclude the existence of a parallel monetary economy with indirect exchanges, as evidenced by the circulation in Mesopotamia of precious metals in the form of unminted bullion. As Goldsmith (1987, p. 11) observed in Chapter 2 on 'The Financial Systems of the Ancient Near East', "[t]hese countries and millennia were essentially nonmonetised, though metals were used to a limited extent in compensatory payments and in the incurrence and discharge of debts, but were never coined." The absence of a proper monetary system meant that the economies of the ancient Near East were organised primarily through administrative redistribution rather than market exchange.

In contrast, the credit theory of money regards the unit of account as the defining feature of a monetary system. Therefore, a system for accounting for barley rations constitutes a monetary system even though barley rations

are not used as a medium of exchange, as emphasised by Ingham (2004). By conflating credit with money, this theory extends the concept of a monetary system to encompass virtually any interpersonal credit relationship and treats any measure of value as a monetary unit.

Puzzle of the rate-of-return dominance of money

As money generally earns little or no interest in the real economy, credit dominates money in terms of expected returns. The puzzle of the rate-of-return dominance of money is the question of why individuals hold money rather than credit despite its lower return. This puzzle has been a prominent focus of economists, as evidenced by Hicks (1935, p. 5), who stated the following:

> What has to be explained is the decision to hold assets in the form of barren money, rather than of interest- or profit-yielding securities. ... This, as I see it, is really the central issue in the pure theory of money.

More recently, Hellwig (1993, p. 218-219), in his Presidential Address to the European Economic Association, asked the following:

> Why does 'worthless' fiat money have a positive value in exchange against goods and services when there are other assets whose own rates of return in each period exceed the own rate of return on money? ... If there is an asset whose own rate of return exceeds the own rate of return on money in each period with probability one and if the asset has the same marketability properties as fiat money, then there exists no rational expectations equilibrium in which fiat money has a positive real value.

The rate-of-return dominance of money is a puzzle for proponents of the credit theory of money but not for proponents of the economic good theory. If money and credit are regarded as being of the same nature and serving the same purposes, it is difficult to explain why economic agents would choose to hold unremunerated money rather than interest-bearing credit. By analogy, if one confuses salt and sugar, it is puzzling why a cook would sometimes use the more expensive sugar rather than the less expensive salt. However, if one understands that salt and sugar have different natures and serve different purposes, the use of sugar is not puzzling, even if it is more expensive than

salt is. Holding unremunerated money is not puzzling once one recognises that "[m]oney is unique [among classes of financial assets] because it is the only financial asset that endows its holder with a predefined value that is available on demand" (Howden, 2023, p. 164).

It is also worth noting a significant change in the recent conduct of monetary policy that challenges the rate-of-return dominance of money. Central banks conduct their monetary policy by keeping interbank money market rates close to their policy rate. Until a few years ago, central banks controlled money market rates by adjusting the terms on which commercial banks borrowed central bank money. Commercial banks were used to pay an interest rate close to the policy rate to borrow money from the money market or the central bank. Holding money did not earn interest, only lending money did (see Panel (a) in Figure 1). Interest was the remuneration for credit operations, not for money holding. This clear distinction between money and credit aligned with the economic good theory of money.

However, the holding of central bank money has recently started to earn interest. Since the global financial crisis of 2008, the monetary base has increased significantly in many jurisdictions following the quantitative easing operations of central banks. With ample central bank money available, the interest rate in the money market fell to almost zero. When the time came for central banks to raise interest rates, they could have reduced the supply of central bank money, creating scarcity conditions similar to those before the global financial crisis. This would have encouraged commercial banks to borrow central bank money at interest, as they always have done. However, this would have required central banks to reduce their balance sheets, thus undertaking quantitative tightening on a scale broadly equivalent to the previous easing. Anticipating that quantitative tightening would not allow interest rates to rise in time without destabilising the financial markets, central banks started controlling money market interest rates by paying interest on holdings of central bank money. Consequently, the policy rate no longer reflects the rate at which banks borrow money from the money market and the central bank; it now reflects the rate they receive for holding central bank money (see Panel (b) in Figure 1).

Although this transition was driven purely for pragmatic reasons, namely, to enable the central bank to raise interest rates despite abundant central bank money, it is a move perfectly consistent with the credit theory of money. If

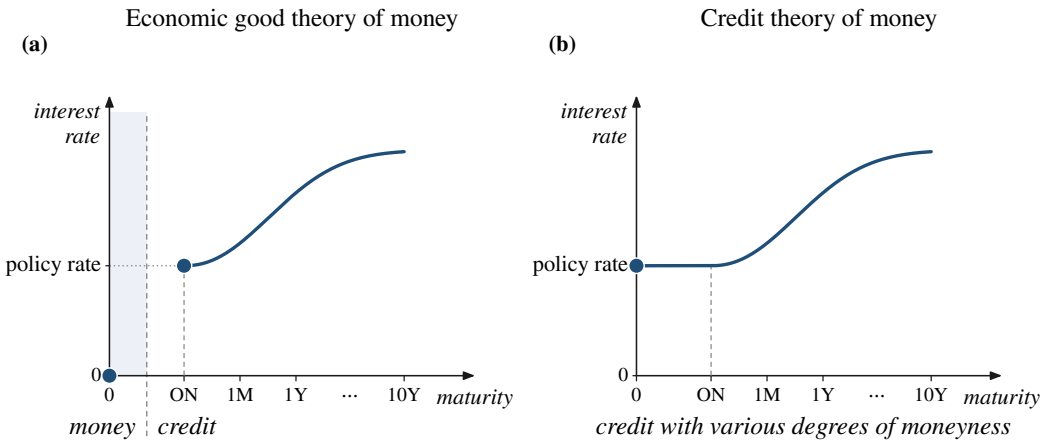


Figure 1: Yield curve in the economic good theory and credit theory of money

money is credit, why does it not earn interest? However, this raises several issues. Since central bank money earns the policy rate, should bank deposits not earn at least that rate? And if they did, would bank lending rates not increase significantly? It is worth noting here that central banks and regulators have taken all necessary measures to prevent this from occurring. Examples of this include refusing to grant banking licences to narrow banks and prohibiting interest payments on stablecoins. These measures prevent the policy rate from being passed on to the bank deposit rate and reflect the conceptual tensions between the modern credit theory and the historical dichotomy between money and credit, which conceives of interest solely as remuneration for credit.

Challenge of the uniformity of money

The quest for the uniformity of money provides strong support for the economic good theory of money. The uniformity of money refers to the principle that different forms of money, such as central bank-issued banknotes and commercial bank-issued deposits, are exchangeable at par and are used interchangeably and without distinction, regardless of their issuer (Bernhard and Haene, 2025). The challenge of achieving monetary uniformity in the current monetary system stems from the coexistence of two distinct forms of money. The first is central bank money, which consists of banknotes and central bank reserves (i.e., sight deposits held by commercial banks). The second is commercial bank money, which consists of deposits issued by commercial banks and redeemable in central bank money. Central bank money is unredeemable fiat money and

therefore accords with the economic good theory of money. Commercial bank deposits, in contrast, are claims on the issuing bank and therefore conform to the credit theory of money. As such, they inevitably expose their holders to the credit risk of the issuing bank. If that risk materialises, the deposits may default, resulting in a loss of money.

Monetary uniformity is an inherent feature of the concept of a 'commonly accepted medium of exchange', which lies at the heart of the economic good theory of money. For a medium of exchange to be accepted universally and without questions asked, it must be uniform: one unit must always be interchangeable with another at face value. Credit claims, in contrast, are inherently nonuniform because the circumstances under which they may not be honoured differ from one claim to another. As a result, credit instruments are valued differently according to the risk they embody. If the credit theory of money is correct and money is fundamentally a form of credit, then the loss of money arising from the materialisation of credit risk must be regarded as a normal and unavoidable feature of the monetary system. There is no such thing as risk-free credit: every credit claim is subject to the possibility of default, and money, if it is merely a credit claim, cannot logically be exempt from that risk.

To preserve monetary uniformity, however, central bank money and commercial bank money must be treated as economically equivalent. They must be fully convertible into one another at par, allowing payments to be settled in either form without distinction or concern about creditworthiness. Achieving this equivalence is difficult precisely because bank deposits are credit claims and therefore inherently risky. The institutional architecture of the current monetary system addresses this problem by socialising the credit risk embedded in commercial bank money, thereby making bank deposits appear to possess the characteristics of an economic good rather than those of a credit claim. This architecture includes the central bank's role as a lender of last resort, deposit insurance schemes, and, in exceptional circumstances, government support or bailouts for distressed banks. These mechanisms reveal a fundamental social and political unwillingness to accept losses of money arising from bank credit risk, as well as the widespread perception of money as an economic good rather than as a form of credit. In contrast, the liabilities of nonfinancial corporations, such as corporate bonds and other debt instruments, receive no comparable protection. Losses on these instruments are generally

accepted as a normal consequence of credit relationships.

4 Controversy about the double-spending of money

A deposit is something given another for safekeeping.

Ulpian

The money placed in the custody of a banker is, to all intents and purposes, the money of the banker, to do with it as he pleases.

Lord Cottenham

The emergence of tokenisation and money tokens has been made possible by advances in cryptography that prevent double-spending of tokens in the absence of a central account holder. In legacy custody systems, transfers between depositors are handled by a custodian or depository who maintains a central register of each depositor's holdings. The key cryptographic innovation is the development of encrypted, distributed, digital ledger systems that eliminate the need for a central custodian and ensure that a token cannot be held by more than one holder at the same time.

At first glance, the principle of not spending tokens twice appears to be sound and unquestionable. However, the question of the double-spending of money has a long history, and the current monetary system is actually based on double, or even multiple, spending of money. Ultimately, the issue is related to the elasticity – or inelasticity – of the money supply. An elastic supply is typically predicated on the possibility of spending the same unit of money more than once. Conversely, the no double-spending principle of cryptocurrencies such as Bitcoin makes their supply inelastic, which may be viewed as a strength by some or a weakness by others. For instance, BIS (2025) identified three key features for which money must be fit for purpose: singleness, elasticity and integrity. The authors asserted, like Selgin (2015), that money must be elastic

to fulfil its role as the mainstay of the monetary system, which suggests that cryptocurrencies lack the elasticity needed to become the money of the future.

The issue of double-spending of money is related to controversies at both the individual and aggregate levels. At the individual level, the issue of double spending relates to the contractual relationship between depositors and custodians: should custodians allow one depositor to spend the money belonging to another depositor? At the aggregate level, the issue of double spending addresses the macroeconomic effect of an elastic money supply: is an elastic money supply beneficial to economic development? In this section, we do not discuss the macroeconomic arguments for or against an elastic money supply. We focus solely on the controversy surrounding the contractual relationship between depositors and custodians. This allows us to classify money tokens according to whether their issuance makes the money supply elastic or not.

Should custodians allow one depositor to spend the money belonging to another depositor? This question addresses custodians' obligations towards depositors. What are the obligations of custodians? To address the controversy about double spending, we follow Huerta de Soto (2009) and begin by outlining the differences between regular and irregular deposits. We then discuss the arguments against double spending raised by classical Roman jurisprudence and the arguments in favour of it raised by Anglo-Saxon common law.

4.1 Regular and irregular deposits

Are the custodian's obligations different for regular and irregular deposits? Deposits are distinguished according to whether the good deposited is a *specific* good, such as a painting or a car, or a *fungible* good, such as oil or money. The deposit of a specific good is called a *regular* deposit, and the deposit of a fungible good is called an *irregular* deposit. In the case of a regular deposit, the custodian has the obligation to return to the depositor the very same – specific – goods deposited. If a Rembrandt painting is deposited with a gallery, that gallery cannot return a Van Gogh painting, nor can it return another Rembrandt painting. It must return the very same painting.

In the case of an irregular deposit, however, the custodian has the obligation to return to the depositor the same quantity and quality of the goods deposited but not necessarily the very same units of the goods deposited. This is because fungible goods are generally mixed indistinguishably with other goods of the same kind. If you deposit 100 barrels of Brent crude oil with a warehouse, it is

likely that your barrels will be stored in a large tank alongside those of other people, meaning that they can no longer be distinguished from one another. The warehouse is required to return 100 barrels of oil of the same quality as the oil you deposited but not necessarily the same molecules that you deposited. Since the depositor does not receive the same unit of the deposited good, he cannot be sure and has no way of checking whether the custodian has used it.

The controversy is whether the fungible nature of the goods deposited removes the safekeeping obligation of the custodian. Should the custodian be entitled to use the deposited goods because the depositor cannot see and control whether the goods have been used or not? Does the safekeeping obligation of the custodian depend on whether the deposited goods are fungible or not?

The legal controversy about the double-spending of money resonates with the controversy about the nature of money described in the previous section. The question of the safekeeping obligation of the custodian of money is inextricably linked to the question of the nature of money itself. The prohibition of double spending in classical Roman law supports the *economic good theory of money*: the distinction between money and credit echoes the distinction between deposits and loans. The acceptance of double spending in Anglo-Saxon law supports the *credit theory of money*: the confusion between money and credit echoes the confusion between deposits and loans.

4.2 Prohibition of double spending in classical Roman jurisprudence

According to classical Roman jurisprudence, the purpose of a deposit contract is the safekeeping of the deposited good, which means that the custodian is not allowed to dispose freely of the deposited good. When banks comply with the classical Roman principles, the banking system is said to be a full or 100% reserve requirement system. This is because banks hold all the money deposited with them for safekeeping and do not allow the money held by depositors to be double spent.

In their analysis of the legal principles governing the deposit contract, Roman lawyers emphasised that the deposit contract and the loan contract have different purposes. The purpose of the deposit contract is the safekeeping of the deposited good (money). In the *Digest*, a subpart of the *Corpus Juris Civilis* issued by Byzantine emperor Justinian in the 6th century A.D., Ulpian

(3rd century A.D.) stated the following:

> A deposit is something given another for safekeeping. It is so called because a good is posited [or placed]. The preposition *de* intensifies the meaning, which reflects that all obligations corresponding to the custody of the good belong to that person.

In contrast, the purpose of the loan contract is to transfer the availability of the loaned good to the borrower so that he can use it for a certain period. The distinction between the deposit contract and the loan contract clearly appears in Ulpian, who continued:

> Whenever bankers are declared bankrupt, usually addressed first are the concerns of the depositors; that is, those with money on deposit, not those earning interest on money left with the bankers. So, once the goods have been sold, the depositors have priority over those with privileges, and those who received interest are not taken into account – it is as if they had relinquished the deposit.
>
> . . . those who have received interest from the bankers on money deposited will not be dealt with separately from the rest of the creditors; and with good reason, since to loan is one thing and to deposit is another.

A deposit cannot be equated with a credit transaction because it lacks the essential purpose of credit, which is the exchange of goods over time. Indeed, the depositor does not grant credit because he constantly holds his money, since his deposit can constantly be used to settle transactions at any time. The bond holder, on the other hand, relinquishes control over money in favour of the bond issuer, to whom he grants credit.

This difference in purpose implies a legal difference in the transfer of ownership and availability of the good. The purpose of the loan contract is to transfer the full availability of the loaned good from the lender to the borrower for the duration of the loan. In contrast, the purpose of the deposit contract can only be realised if the full availability of the deposited good remains in favour of the depositor. The fundamental element of the deposit contract is the obligation of the custodian to always maintain the full availability of the good in favour of the depositor. If the banker uses the money deposited with him, the money

is no longer available to the depositor, which is precisely the purpose of the deposit contract.

In the *Corpus Juris Civilis*, Gordianus (3rd century A.D.) emphasised that depositaries were not entitled to use the money deposited with them but were obliged to keep it safe for the depositor. He stated that the custodian who makes use of the deposit entrusted to him is guilty of theft by misappropriation:

> ... he who knowingly and willingly uses a deposited good for his own benefit, against the will of the owner, also commits the crime of theft.

In summary, classical Roman jurisprudence condemns the double-spending of money. The no double-spending principle of cryptography is compatible with it.

4.3 Acceptance of double spending in Anglo-Saxon common law

In contrast, the Anglo-Saxon common law tradition treats the deposit contract of money as a loan contract, giving the custodian free rein over the money deposited. When banks comply with the Anglo-Saxon principles, the banking system becomes a fractional-reserve system because banks hold only a fraction of the money deposited for safekeeping, while the other part is spent.

Throughout history, bankers have repeatedly failed to maintain the full availability of the money deposited with them in favour of depositors and have succumbed to the temptation to use their depositors' money for their own benefit. Over the centuries, this practice gave rise to numerous lawsuits in which depositors sued their bankers for theft and fraud in the exercise of their safekeeping obligation. The Roman jurisprudence described in the previous section bears witness to these legal disputes that pitted depositors against their bankers and shows that Roman lawyers ruled in favour of depositors against bankers. In continental Europe, many court decisions upheld the obligation of bankers to maintain a 100% reserve requirement in accordance with classical Roman legal principles.[6] However, the British judicial system began to accept and justify the double-spending of money in the first half of the nineteenth

[6]For example, Huerta de Soto (2009, p. 11) stated that as late as 1927, "the Court of Paris convicted a banker for the crime of misappropriation for having used, as was the common practice in banking, funds deposited with him by a client."

century on the basis that deposits were considered loans made by depositors to bankers.

In the case of *Sims v Bond* (1833), the Queen's Bench Division (the High Court of Justice) affirmed the following:

> Sums which are paid to the credit of a customer with a banker, though usually called deposits, are, in truth, loans by the customer to the banker.

The judges argue that the depositor is actually making a loan to the banker. They nevertheless consider the term *deposit* to be misleading. The phrase "though usually called deposits" reveals their understanding that a deposit is not a loan made by the depositor to the banker. However, rather than condemning the banker for using the deposited money for their own purposes on the grounds that the deposited money is not being lent, the court equates the deposit with a loan while acknowledging that the term *deposit* is unfortunate. Therefore, bankers who spend money deposited with them for their own purposes are not guilty of theft. In the judgement of *Foley v Hill* (1848), Judge Lord Cottenham stated the following:

> The money placed in the custody of a banker is, to all intents and purposes, the money of the banker, to do with it as he pleases. He is guilty of no breach of trust in employing it. He is not answerable to the principal if he puts it into jeopardy, if he engages in a hazardous speculation; he is not bound to keep it or deal with it as the property of his principal, but he is, of course, answerable for the amount, because he has contracted, having received that money, to repay to the principal, when demanded, a sum equivalent to that paid into his hands.

British jurisprudence institutionalises the double-spending of money deposited with banks. This practice has spread around the world to shape the current monetary system of fractional-reserve banking, which is increasingly orchestrated and backed by a central bank as a lender of last resort. It is important to note that the principle of no double spending, on which certain money tokens are based, is conceptually at odds with the current monetary system.

5 How do different types of money fit in with competing ontologies?

Having discussed the competing views regarding the origin, nature and possibility of double-spending of money, we can now classify the various forms of money according to the ontological categories they represent. The things used as money have arisen either spontaneously through market exchange processes or through state decree. They may take the form of economic goods or credit claims, and they may or may not possess the capacity for double spending.

This exercise involves an important conceptual qualification. As noted in the Introduction, controversies about the ontology of money transcend its particular form, i.e., its condition. An ontology of money seeks to explain what money is in essence, irrespective of the form it takes. For example, the credit theory holds that money is fundamentally credit, even when it takes the form of gold. In presenting his theory of credit, Macleod (1889, p. 72) wrote:

> Gold and Silver Money may, therefore, be justly termed Metallic Credit.

This statement may appear surprising, since precious metals are neither claims on an issuer nor subject to credit risk. Nevertheless, Macleod considered gold fully consistent with the credit theory of money, even though identifying a commodity with a credit relationship is conceptually challenging. Similarly, the state theory of money maintains that money is a creation of the state even when it consists of gold, as illustrated by quotations from Michell-Innes in Sections 2.2 and 2.3. However, despite treating gold as compatible with their state and credit theories of money, both Macleod and Michell-Innes viewed the establishment of a gold-based monetary system with disfavour, implicitly acknowledging that commodity money sits uneasily with their theoretical frameworks. In this sense, competing ontologies are not neutral or indifferent with respect to the form of money, despite their aim to capture money regardless of its form. For the purposes of the analysis that follows, we therefore associate particular monetary forms with the ontology that most naturally characterises them. Thus, we treat, for example, gold as neither a credit nor a state-created money, notwithstanding the fact that Macleod and Michell-Innes would interpret gold money as both a credit and a creation of the state.

Nature of money			Origin of money: Natural law	Origin of money: State theory
Economic good	No double spending		**A1** • precious metal • cryptocurrency • note/deposit/stablecoin redeemable in precious metal or cryptocurrency and fully backed by it	**B1** • state-issued fiat money: note/deposit/CBDC • deposit/stablecoin redeemable in state-issued fiat money and fully backed by it
Credit Claims to goods	Double spending		**A2** • note/deposit/stablecoin redeemable in A1 but partially backed by A1 and partially by other assets	**B2** • deposit/stablecoin redeemable in B1 but partially backed by B1 and partially by other assets
Credit Claims to goods	No double spending		**A3** • stablecoin redeemable in and backed by A2 (and possibly A1)	**B3** • stablecoin redeemable in and backed by B2 (and possibly B1)

Table 1: Origin, nature and double-spending of types of money

5.1 Natural law and economic good theory of money

Forms of money that conform to the natural law and economic good theories are listed in cell A1 of Table 1. The most prominent and historically significant example is precious metals. Precious metals are economic goods rather than claims on any issuer since they are not redeemable in anything else. According to the regression theorem, commodities – and precious metals in particular – gradually emerged as money through the process of market exchange, independent of the state. Indeed, precious metals served as media of exchange long before states began minting coins. Moreover, although states may mint coins, the quantity of metal available for monetary use is ultimately determined by natural availability and market forces rather than by political decisions.

Originally, the economic good theory of money was equated with a commodity, or metallist, theory of money, reflecting the historical prominence of precious metals as money. However, the concept of money as an economic good is not limited to tangible commodities. Intangible assets that function as media of exchange may also satisfy the economic good theory. Cryptocurrencies such as Bitcoin are economic goods and may be used as a medium of

exchange. Because they do not constitute redemption claims on any issuer, they are consistent with the economic good theory of money. Moreover, having emerged spontaneously rather than by state decree, cryptocurrencies also conform to the natural law of money. As argued above, their emergence as a monetary phenomenon was probably facilitated by the existence of state-issued fiat money, and they may therefore be regarded as an unintended creation of the state. Nevertheless, this indirect influence does not change the fact that cryptocurrencies were not created by the state and are not under its control.

Banknotes, deposits, and stablecoins that are redeemable in, and fully backed by, precious metals (or cryptocurrencies) are also consistent with the natural law and economic good theory of money. It is true that such money substitutes involve counterparty risk, since the issuer or custodian may fail to hold sufficient precious metals (or cryptocurrencies) to redeem all outstanding claims. Nevertheless, if they are fully backed by the asset in which they are redeemable on demand, these substitutes faithfully represent the underlying asset, share its economic characteristics, and therefore remain consistent with the economic good theory of money. This is the case for banknotes, deposits, and stablecoins issued by institutions operating in accordance with the principles of classical Roman jurisprudence, which require 100 percent reserve holding and thereby prevent the double-spending of deposited money.

If fully backed banknotes and deposits accurately represent their underlying asset, one may ask why they were introduced in the first place. It is therefore important to emphasise the advantages that banknotes and deposits offer over precious metals and the advantages that deposits offer over banknotes. As coins are vulnerable to theft and are not practical for remote payments, people have deposited their coins with banks, thereby making access to money in the form of banknotes or deposits more convenient. Making a payment with a banknote or through a bank transfer is considerably simpler than transporting physical metal. Deposit banks were established to provide depositors with greater security against theft and to facilitate easier access to payments. The original purpose of banks was therefore to make access to money more convenient. Keeping this in mind will be useful when discussing the forms of state-issued money below to understand why issuing CBDCs, which are as convenient as deposits, poses a threat to banks.

It is worth noting that some confusion may arise from the different meanings assigned to the term ‘reserve’ in the banking and stablecoin literature. In

the banking literature, 'full-reserve' and 'fractional-reserve' refer to the amount of base money held by a bank relative to its customers' deposits. Under a natural law money system, base money consists of precious metals or privately issued fiat money, as imagined by Hayek (1976), whereas under a state theory of the money system, it consists of state-issued fiat money. A full-reserve deposit is therefore a deposit fully backed by the base money in which it is redeemable. A fractional-reserve deposit is only partially backed by base money, with the remainder being backed by other asset positions, such as loans or securities. In the stablecoin literature, however, 'reserve' refers to the total pool of assets backing the stablecoin, regardless of whether these assets consist of base money or credit positions such as securities. Consequently, the statement that a stablecoin is fully backed by its reserve assets does not imply that it is fully backed by base money in the sense of a full-reserve deposit. Rather, it means that the value of its reserve assets is at least equal to the stablecoin's redemption value, a condition analogous to the solvency of a fractional-reserve bank. Therefore, determining whether a stablecoin conforms to the economic good or credit theory of money requires examining the nature of its backing: if it is backed exclusively by base money, it conforms to the economic good theory; if it is backed, wholly or partly, by credit positions such as securities or fractional-reserve bank deposits, it conforms to the credit theory.

5.2 Natural law and credit theory of money

Forms of money that conform to the natural law and credit theories are listed in cells A2 and A3 of Table 1. These include banknotes, deposits and stablecoins denominated and redeemable in precious metals or cryptocurrencies but only partially backed by those assets, with the remainder backing consisting of other assets such as loans or securities.

As these forms of money are denominated and redeemable in a base money that complies with the natural law, they themselves fall within this category. However, since the issuer does not hold the deposited money in custody but instead invests, lends, or otherwise disposes of it, these claims are no longer fully representative of base money but are credit relationships. People accept these fractional-reserve banknotes and deposits as a final means of payment insofar as they have confidence in the solvency of the issuer, that is, insofar as the value of a bank's assets exceeds its liabilities. Because they are backed by assets other than the asset in which they are redeemable, these instruments

entail credit risk and conform to the credit theory of money.

Banks issue fractional-reserve banknotes and deposits by extending credit through loans or the purchase of securities. This process increases the quantity of deposits in circulation, making the money supply elastic to credit demand. This expansion is made possible because the base money owned by the original depositor simultaneously serves as the basis for newly created credit, resulting in the effective double use of the same base money.

Historically, commercial banks were used to issue banknotes redeemable in precious metals. Over time, however, central banks have acquired a monopoly over banknote issuance. In the United Kingdom, for example, the Bank Charter Act, enacted by Prime Minister Robert Peel in 1844, granted the Bank of England the exclusive right to issue banknotes. These banknotes were redeemable in precious metals until legal tender legislation removed the obligation of redemption, turning them into fiat money. Commercial banks, however, have retained the right to issue deposits, although these are now redeemable in central bank money rather than precious metals.

Stablecoins can likewise be classified according to whether they allow the double-spending of the base money in which they are denominated. Stablecoins that are redeemable in precious metals or cryptocurrencies but only partially backed by those assets, with the remainder invested in securities or other financial assets, operate on fractional-reserve principles. Like fractional-reserve banks, they expand the quantity of money in circulation (provided, of course, that stablecoins are considered money and are used to settle day-to-day transactions). Such stablecoins therefore belong in cell A2.

In contrast, stablecoins that are fully backed by the asset in which they are redeemable operate on full-reserve principles because they do not reuse the assets deposited by stablecoin holders. Conceptually, this includes stablecoins redeemable in fractional-reserve bank deposits, which are themselves redeemable in precious metals, and fully backed by those deposits (cell A3 in Table 1). Although such an arrangement is not feasible today because bank deposits are no longer convertible into precious metals, it remains analytically useful, as its analogue with state-issued fiat money is common in modern financial systems. While these stablecoins do not themselves engage in the double use of the deposits backing them, the deposits themselves originate from the fractional-reserve creation of money by commercial banks. Consequently, such stablecoins inherit the monetary characteristics of their backing assets and

therefore conform to the credit theory of money, even though the stablecoin issuer itself operates on a full-reserve basis and does not increase the amount of money in circulation.

5.3 State and economic good theories of money

Forms of money that conform to the state and economic good theories are listed in cell B1 of Table 1. These include the various forms of state-issued fiat money, such as inconvertible banknotes, sight deposits with the central bank, and, potentially, CBDCs.

Legal tender laws were first enacted to reduce the precious metal content of coins and, over time, paved the way for the emergence of fiat money, which has no intrinsic value. Despite lacking intrinsic value, legal tender fiat money is consistent with the economic good theory because it embodies no claim against the central bank, the state or any other party. It is nonconvertible, and its exchange value derives not from a redeemable claim but from economic agents' subjective valuation of its usefulness as a medium of exchange and store of value. The key distinction between legal tender fiat money and cryptocurrency is that the former is issued and controlled by the state, whereas the issuance of the latter is governed by cryptographic protocols operating outside the control of the state.

Although the three potential forms of state-issued fiat money share the same underlying ontology, they differ fundamentally in their accessibility to the public. Banknotes are universally accessible and can be held directly by individuals without restriction. In contrast, individuals typically have no access to sight deposits with the central bank, while access to CBDCs, where available, is generally limited by design. This difference in accessibility reflects a fundamental institutional trade-off: the provision of a form of central bank money that offers the same convenience as commercial bank deposits could undermine the foundations of the fractional-reserve banking system. As discussed in Section 5.1, individuals began depositing precious metal coins with banks because banknotes and deposits provided greater security and greater ease of payment than holding physical coins themselves did. The same logic continues to apply in modern monetary systems: individuals hold deposits with commercial banks because bank deposits offer greater security and convenience than holding banknotes does. However, if central banks were to provide individuals with unrestricted access to sight deposits or CBDCs that matched commercial

bank deposits in terms of convenience and functionality, the primary incentive for maintaining deposits with commercial banks would diminish substantially. Such a development could weaken the role of commercial banks as monetary intermediaries, credit suppliers and deposit issuers, thereby challenging the institutional foundations of the fractional-reserve banking system. Baeriswyl, Reynard, and Swoboda (2024) explained that the unrestricted provision of CBDCs to individuals would entail a transfer of credit risk from depositors to the central bank. When risky bank deposits are converted into CBDCs, the risk borne by depositors does not disappear; rather, it is transferred to the central bank, thereby challenging the ordnungspolitik coherence of the system. To contain the transfer of risk, access to CBDCs is generally limited by design, such as an individual holding ceiling. Imposing a holding limit implies that individuals will not hold all their money as CBDCs but will instead maintain portfolios consisting of both CBDCs and bank deposits. Since CBDCs are economic goods and free of credit risk and bank deposits constitute risky credit claims, the absence of credit risk is likely to make CBDCs particularly attractive as a store of value. This raises the question of whether CBDCs will nonetheless be widely used as a means of payment. Baeriswyl, Boun-My, and Cornand (2026) provided evidence from a laboratory experiment that individuals may indeed prefer to hold CBDCs as stores of value while spending deposits as a means of payment, echoing the mechanism underlying Gresham's law.

Cell B1 also includes deposits and stablecoins that are redeemable in and fully backed by state-issued fiat money. Stablecoins backed entirely by state-issued fiat money are commonly referred to as synthetic CBDCs.

It should also be noted that the payment of interest on state-issued fiat money does not alter its status as an economic good. According to the economic good theory of money, interest is remuneration for a credit transaction rather than an inherent attribute of holding money. As discussed in Section 3.3, the recent payment of interest on central bank money is a move consistent with the credit theory of money. Nevertheless, the payment of interest does not, by itself, transform central bank money into credit, as it neither becomes a claim recoverable from the issuing institution nor derives its value from such a claim. Even when it bears interest, central bank money remains an economic good rather than a claim against its issuer.

5.4 State and credit theories of money

Forms of money that conform to the state and credit theories are listed in cells B2 and B3 of Table 1. The most prominent and economically significant example is deposits denominated and redeemable in state-issued fiat money, but only partially backed by it, with the remainder backing consisting of other assets such as loans or securities. These fractional-reserve deposits constitute the bulk of the money in circulation in the current monetary system. They represent claims on the issuing bank, bear its credit risk and conform to the credit theory of money. As with their counterparts in cell A2, the issuance of fractional-reserve deposits contributes to an increase in the money supply and is based on the principle of double-spending of base money.

Stablecoins operating in this way and partially backed by securities are economically identical to fractional-reserve deposits and also contribute to an increase in the money supply (provided, of course, that stablecoins are considered money and are used to settle day-to-day transactions).

In contrast, stablecoins that are fully backed by fractional-reserve deposits operate on the full-reserve principle because they do not reuse the assets in which they are redeemable (cell B3 in Table 1). While they do not engage in the double use of the deposits backing them, these backing deposits are themselves credit claims on their issuing banks and conform to the credit theory of money. Such stablecoins share the monetary characteristics of their backing assets and therefore conform to the credit theory of money, even though the stablecoin issuer itself operates on a full-reserve basis and does not increase the amount of money in circulation.

6 Conclusion

The question of which tokens are best suited to becoming the money of the future remains contested. The complexity of modern monetary systems often makes it difficult to disentangle the competing arguments surrounding the merits and shortcomings of different forms of money. The historical controversies presented in this paper provide a framework for understanding these debates, clarifying the fundamental characteristics of monetary instruments and the sources of disagreement over their design. While money has evolved alongside technological progress, the underlying debates about its origin, nature and desirable properties have remained remarkably the same. Contemporary dis-

cussions of cryptocurrencies, CBDCs, and stablecoins are therefore best understood as modern manifestations of enduring theoretical disagreements rather than entirely new controversies. As monetary technologies continue to evolve, these debates suggest that the future of money will remain as contentious and dynamic as its past.

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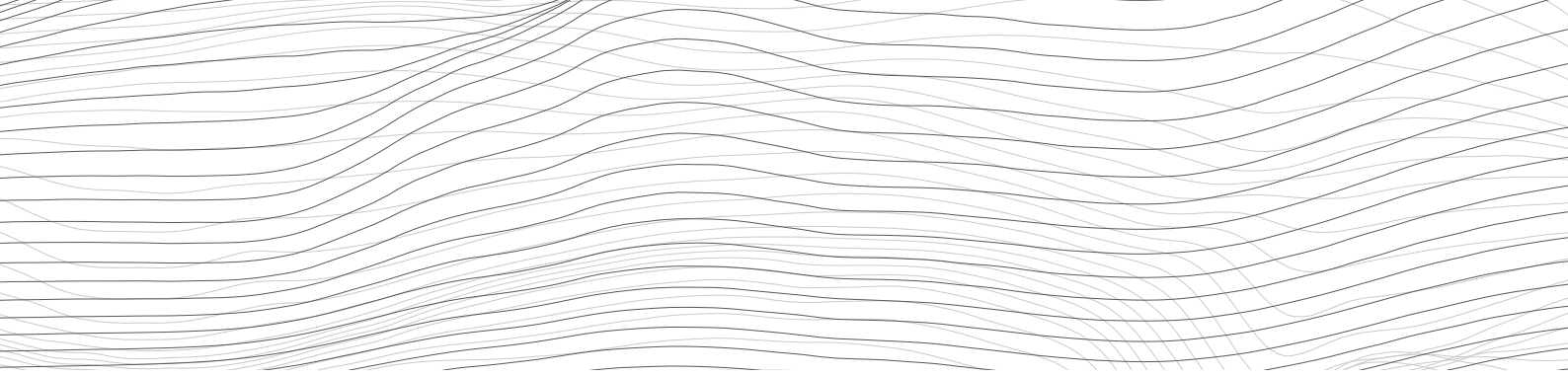
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